

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		K_a of stronger acid would be <u>greater</u> than that for weaker acid as stronger acids have <u>greater dissociation</u>	1			1		
		(ii)	I	Joe's method: rearrangement of $pV = nRT$ (1) $n = 3.5 \times 10^{-3}$ (1) accept alternative method Heledd's method: $n = 3.48 \times 10^{-3}$ or 3.484×10^{-3} (1) must show use of M_r of carbon dioxide to gain mark both answers to appropriate significant figures and both concentrations are the same (ECF from calculations) (1)		1 1		4	3	1
			II	Heledd's method as measurements are to more significant figures / more precise / have higher resolution			1	1		1
			III	would make Heledd's experiment less accurate as the mass released would be much smaller (as H_2 has much smaller M_r than CO_2) (1) will not affect the accuracy of Joe's experiment as same volume of gas would be produced (1)			1 1	2		
		(iii)		pH from 2-6 (1) must attempt reason to gain this mark award (1) for either of following ammonium ion (partially) dissociates to release H^+ ions $NH_4^+ \rightleftharpoons NH_3 + H^+$ do not accept compound/salt/ammonium perchlorate releases H^+	1	1		2		
Question 11 total					2	14	8	24	13	4